

Introduction to Business Analytics

961731 Business Data Analytics

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28/06/2026

Data is the new oil

Clive Humby (2006)

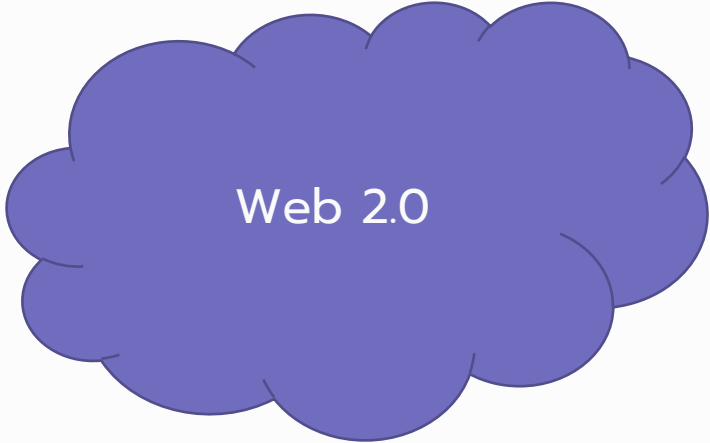
Introduction to Business Data Analytics

- Ubiquity of data opportunity
 - New ways to collect data
- Availability of high volume of data storage
- Accessibility of high performance computing

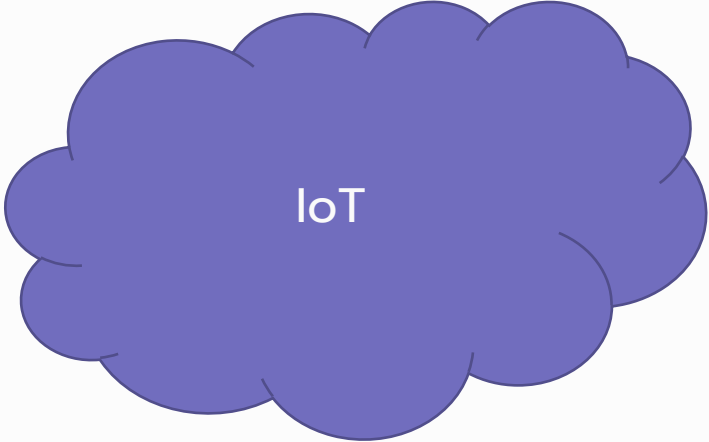


Data-Analytic Thinking

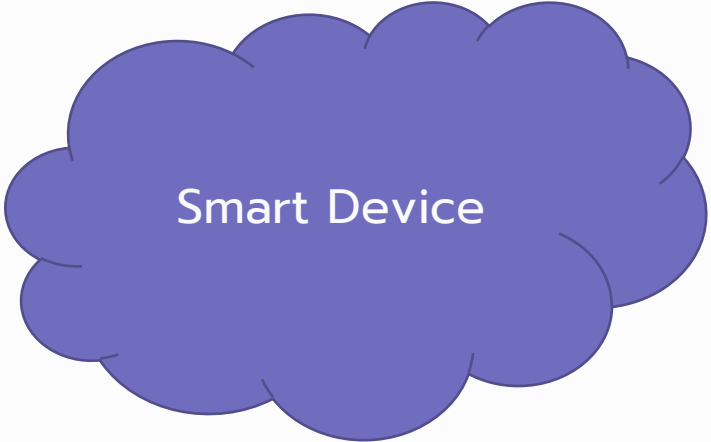
Data Opportunities



Web 2.0



IoT



Smart Device

What guests loved the most:

- "1) Excellent and fast service anywhere in the resort.
- 2) Staff are very friendly and attendant.
- 3) Beautiful sight at the beach and main pool area.
- 4) Excellent kids club facility and activities.
- 5) Spa is amazing.
- 6) Airport shuttle was up to expectation and worth it.
- 7) Attention to details by resort management is clear and well appreciated."

 **Pickles**
wait

!! I really do hope I can visit you for everything!"

uth Korea

to be if all you want is the
e rooms are romantic and the
great.
planned every day to keep
ou finish your vacation book.

Danger of Ignoring Data

- Failing to notice sentiments about the business
 - Aka Opinion Mining

Reviews of Lodge of Four Seasons Golf Resort, Marina & Spa from real guests ⓘ

Provided by verified guests of 

AGODA REVIEWS (2)

BOOKING.COM REVIEWS (360)

5.2/10

Review score

✓ From 2 reviews

Cleanliness 4.0

Location 9.0

Value for money 3.0

Facilities 4.0

Service 6.0

Rating

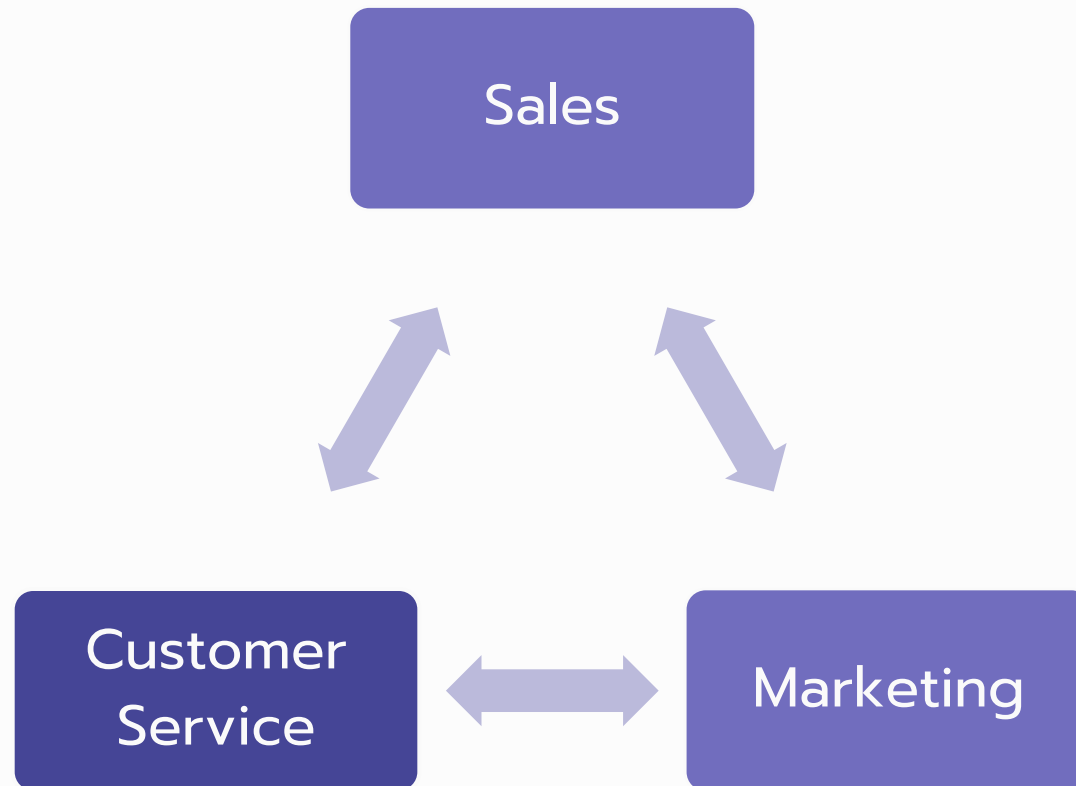
- ☐ 9+ Exceptional (0)
- ☐ 8-9 Excellent (0)
- ☐ 7-8 Very Good (0)
- ☐ 6-7 Good (1)
- ☐ <6 Below Expectation (0)

Time of the year

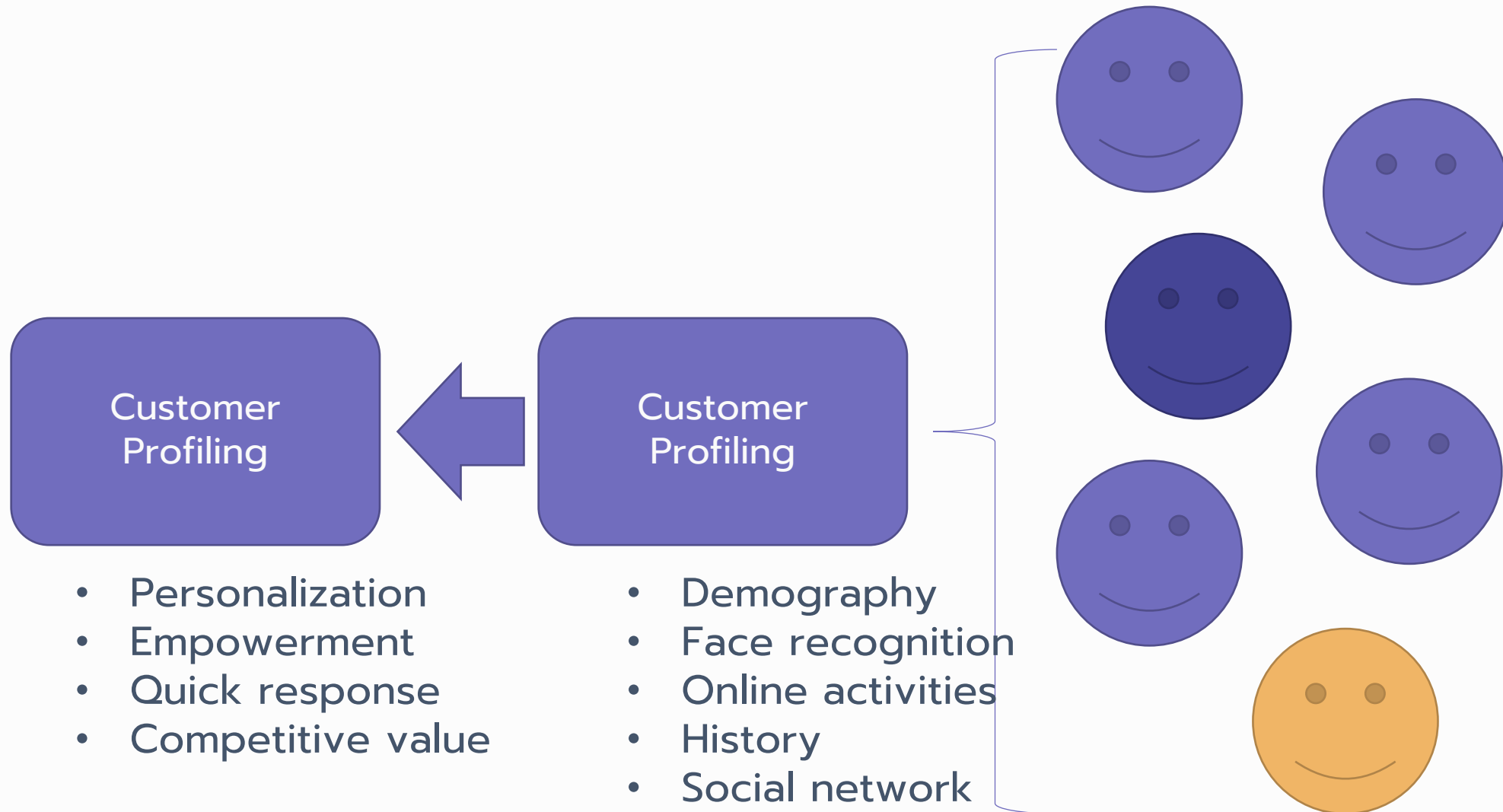
- ☐ Mar - May (0)
- ☐ Jun - Aug (1)
- ☐ Sep - Nov (0)
- ☐ Dec - Feb (0)

Danger of Ignoring Data

- Failing to build strong customer relationships



Danger of Ignoring Data



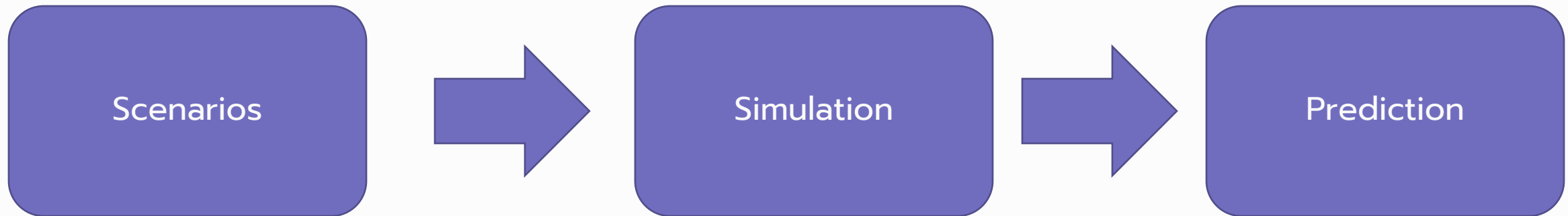
Danger of Ignoring Data

61% of marketing decision makers said they struggled to access or integrate the data they needed last year.

<https://www.thinkwithgoogle.com/marketing-resources/measurement/marketing-analytics-data-challenges->

Danger of Ignoring Data

- Failing to realize the past
 - Utilize of what-if analysis



What if your biggest customer, customers was suddenly to decline by 20%.

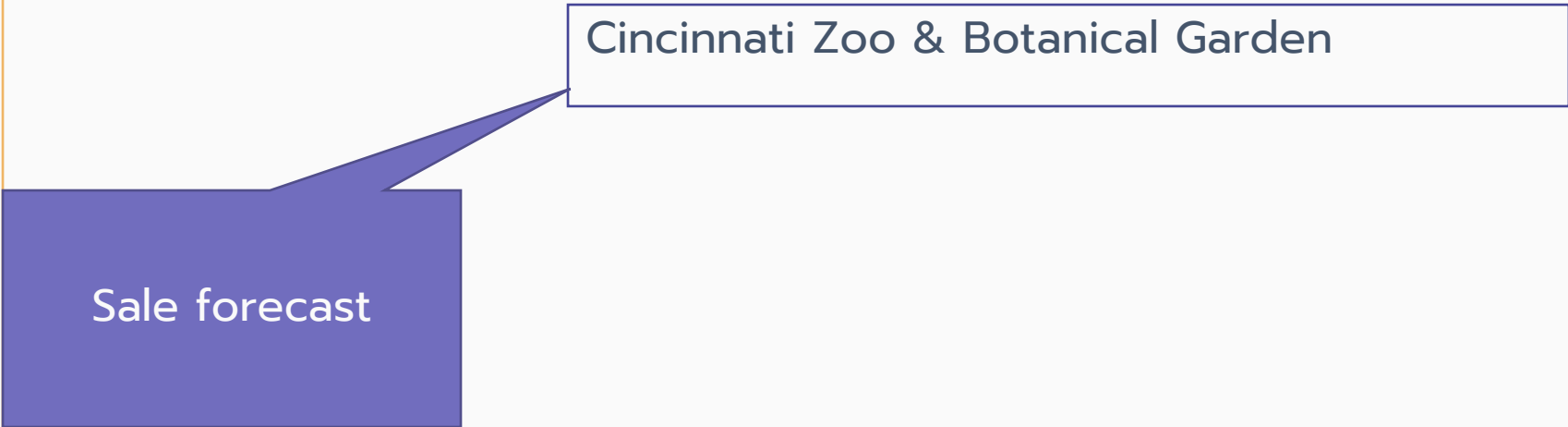
What if your store were to become ten times bigger.

What if a hurricane hit your city.

What if your opponent declined by 20%.

Danger of Ignoring Data

- Failing to realize the past
 - Impact of accurate sale forecasts

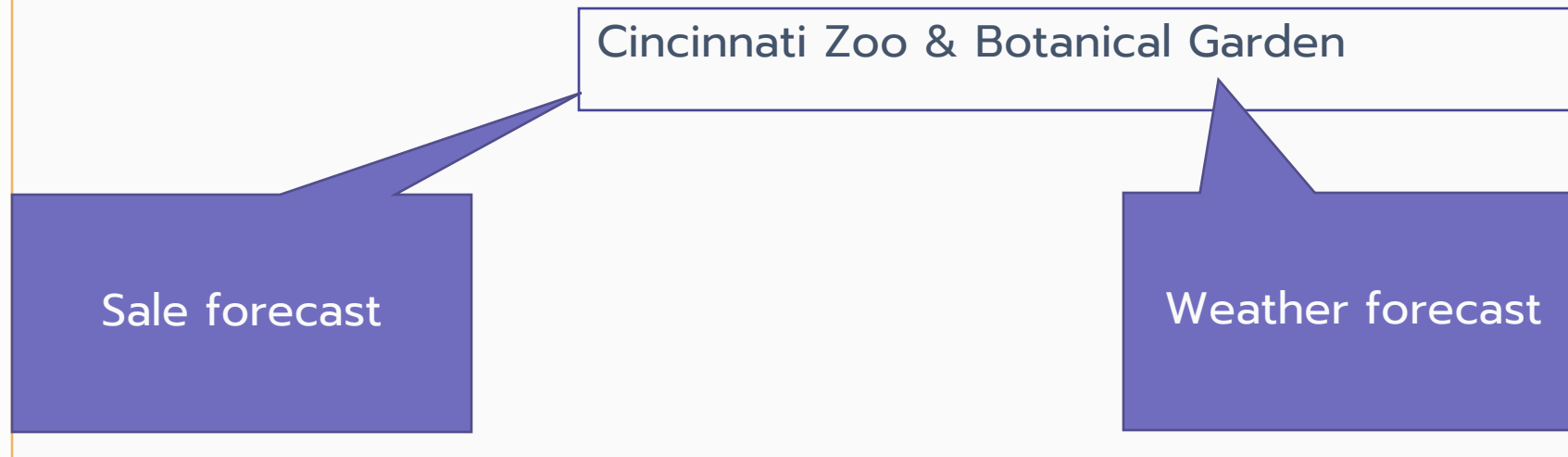


Cincinnati Zoo & Botanical Garden

Sale forecast

Danger of Ignoring Data

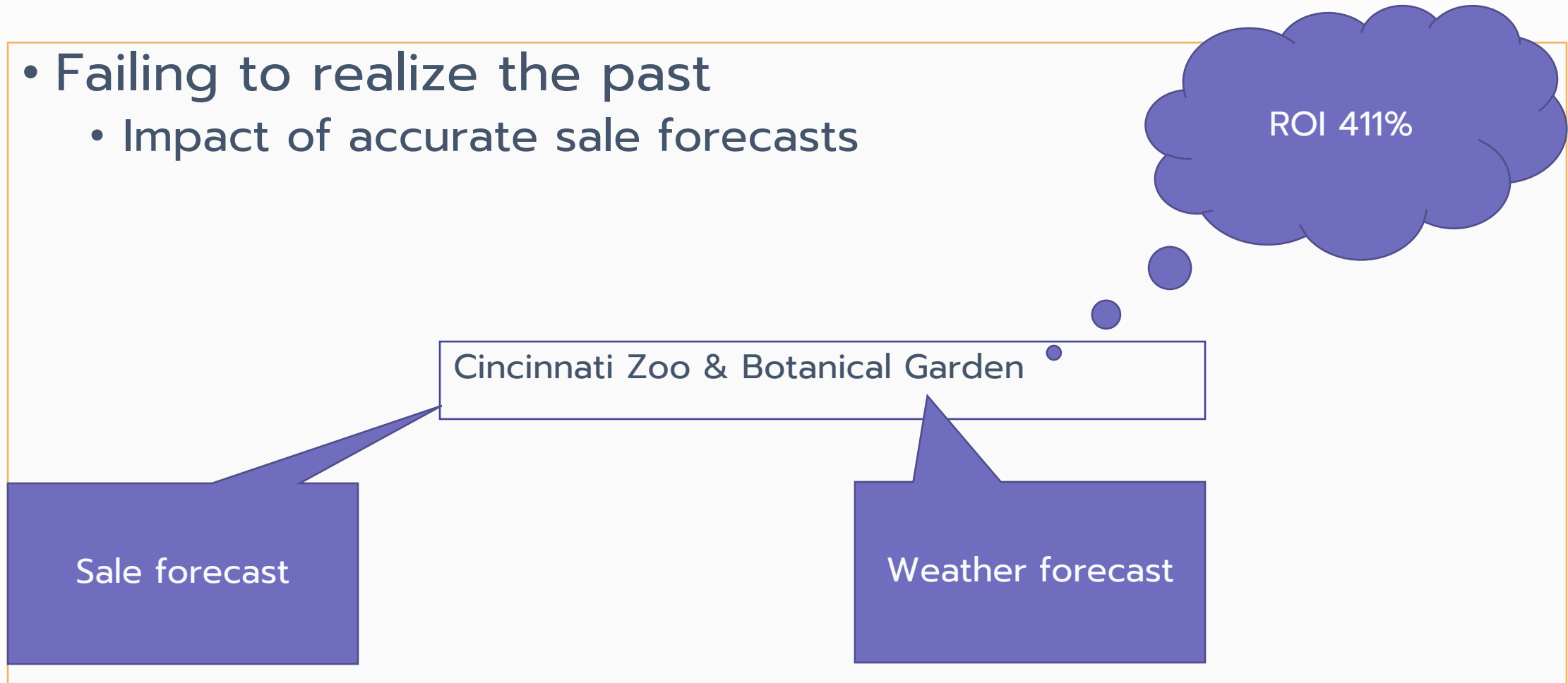
- Failing to realize the past
 - Impact of accurate sale forecasts



IBM Software Business Analytics, "Cincinnati Zoo transforms customer experience and boosts profits," © IBM Corporation 2012

Danger of Ignoring Data

- Failing to realize the past
 - Impact of accurate sale forecasts



Customer targeting

- 4.2% rise in ticket sales

Optimize product mix

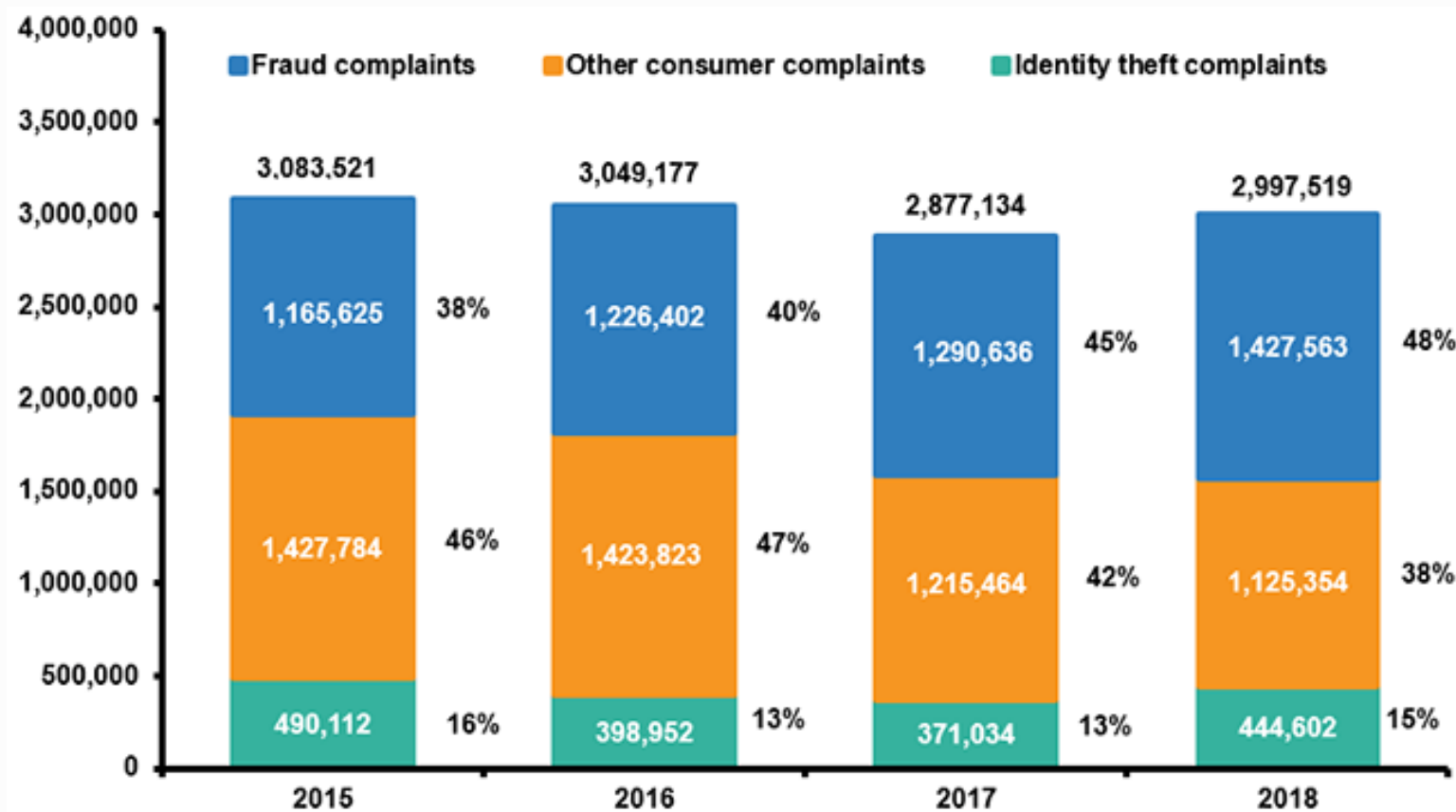
- 25% rise in food revenues

Eliminate ineffective campaigns

- Save 40,000\$ in first year
- 43% reduce in advertising expenditure

Danger of Ignoring Data

- Failing to identify fraud



<https://www.iii.org/graph-archive/96074>

Data-Driven Decision Making

*Practice of basing decisions on the analysis of data,
rather than purely on intuition.*

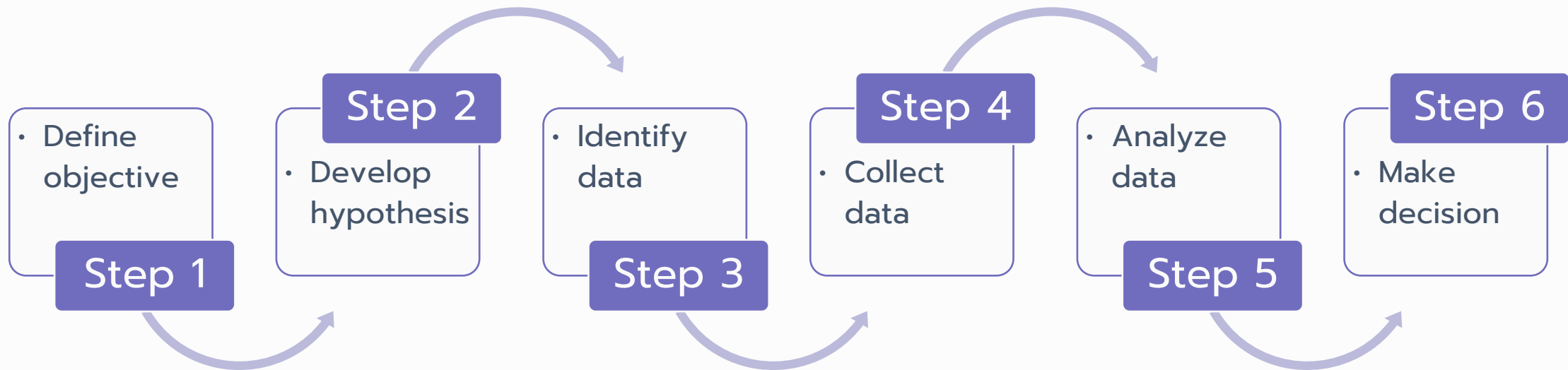
Data-Driven Decision Making

- Five stages to data-driven culture



Data-Driven Decision Making

- Seven steps of data-driven decision making process



Data Processing and Big Data

- Amount of data
 - Is exponentially going up.
- Technology
 - Storage is cheaper and cheaper.
 - High performance computer is accessible.
 - Various ways to gather data.

Evolution of Big Data

Year	Version	Key Developments
1994-2004	1.0	- E-commerce - Web mining (usage mining, structure mining, content mining)
2005-2014	2.0	- Social media mining - Sentimental analysis
2015 ++	3.0	- Streaming platform - Smart device and IOT

Lee, I. (2017). Big data: Dimensions, evolution, impacts, and challenges. Business Horizons, 60(3), 293–303.doi:10.1016/j.bushor.2017.01.004

Data Analytics as Strategic Asset

Consumer Credit Card

1980s

Data Analytics as Strategic Asset

Key characteristics

- 1) Uniform pricing
- 2) No based on data

Credit
limits

Cash
back

Loyalty
point

Consumer Credit Card

1980s

Data Analytics as Strategic Asset

Key characteristics

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Consumer Credit Card

1980s

Early stage problem

- 1) No data

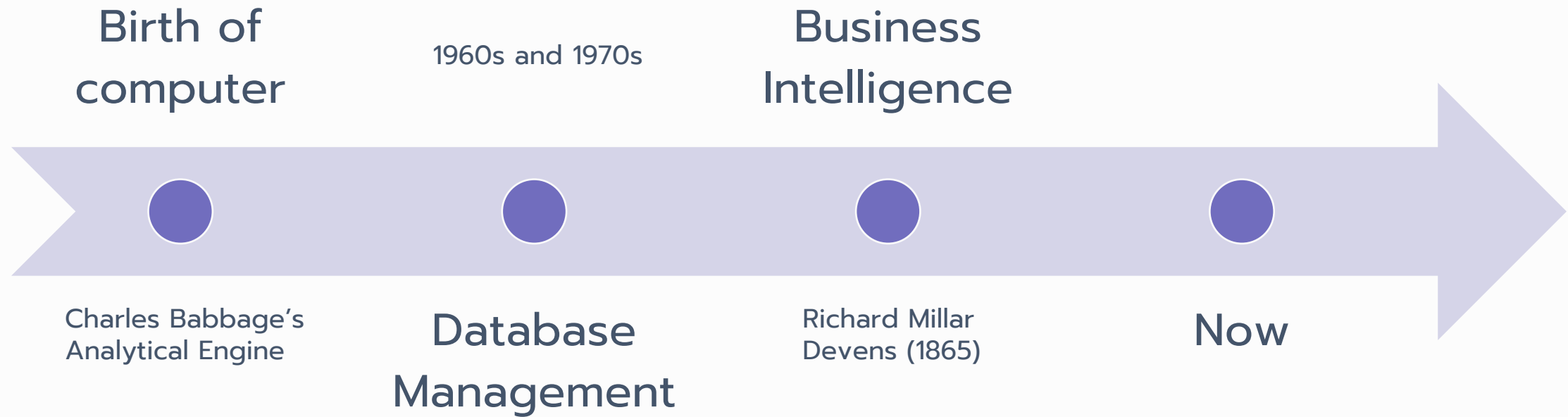


Gather it !!!!

Data Analytics as Strategic Asset

- Data acquisition !!
 - Invest in data
- Conducting experiments (many years)
 - Offer different term at random
 - Charge-off rate goes up from 2.9% → 6%
- After the appropriate model is built, the bank is profitable.
 - Spun off the credit card to be a new company.
 - Capitol One

Evolution of Business Data Analytics



Evolution of Business Data Analytics

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Business Data Analytics VS Business Intelligence

- **Historical vs. Predictive**

- Business Intelligence focuses on using historical data to analyze what has happened in the past assisting decision making process.
- Business Data Analytics uses statistical models and algorithms to understand the trends in data and predict future outcomes.

Business Data Analytics VS Business Intelligence

- **Historical vs. Predictive**
- **Descriptive vs. Prescriptive**
 - Business Intelligence primarily deliver descriptive analytics, i.e., they provide reports, dashboards, and visualizations.
 - Business Data Analytics provides not only predictive analytics but also prescriptive analytics.

Business Data Analytics VS Business Intelligence

- **Historical vs. Predictive**
- **Descriptive vs. Prescriptive**
- **Internal data vs. External data**
 - Business Intelligence deals with structured data from internal business systems like ERPs and CRMs.
 - Business Data Analytics can process not only structured but also unstructured and semi-structured data from various sources, including social media, web logs, sensors, and more.

Ethical Issues in Business Data Analytics

- **Data Privacy**

- Enormous amount of data is collected.
- Risk of violation of data privacy

- **Bias and Fairness**

- Algorithms used in data analytics can unintentionally perpetuate biases present in the data

- **Transparency and Explainability**

- Black boxed operation.

961731

Business Data Analytics

การวิเคราะห์ข้อมูลธุรกิจ

Course Objectives

วัตถุประสงค์กระบวนการวิชา นักศึกษาสามารถ

1. อธิบายหลักการและกระบวนการของการวิเคราะห์ข้อมูลทางธุรกิจ
2. ประยุกต์ใช้หลักการการวิเคราะห์ข้อมูลสำหรับข้อมูลทางธุรกิจต่อกรณีศึกษา

Course Objectives : Students will be able to

1. explain the principle and methodology of data analytics in business.
2. apply the data analytic techniques for business to case studies.

Course Contents

- **Course Contents**

1. Business data-analytic thinking
2. Business problems
3. Business data exploration and summarization
4. Market basket analysis
5. Customer segmentation
6. Customer lifetime value modeling
7. Churn modeling
8. Social media mining

Score and Grading System

- Criteria reference
- Score
 - Midterm 40%
 - Final 40%
 - Assignment 20%

Text Book

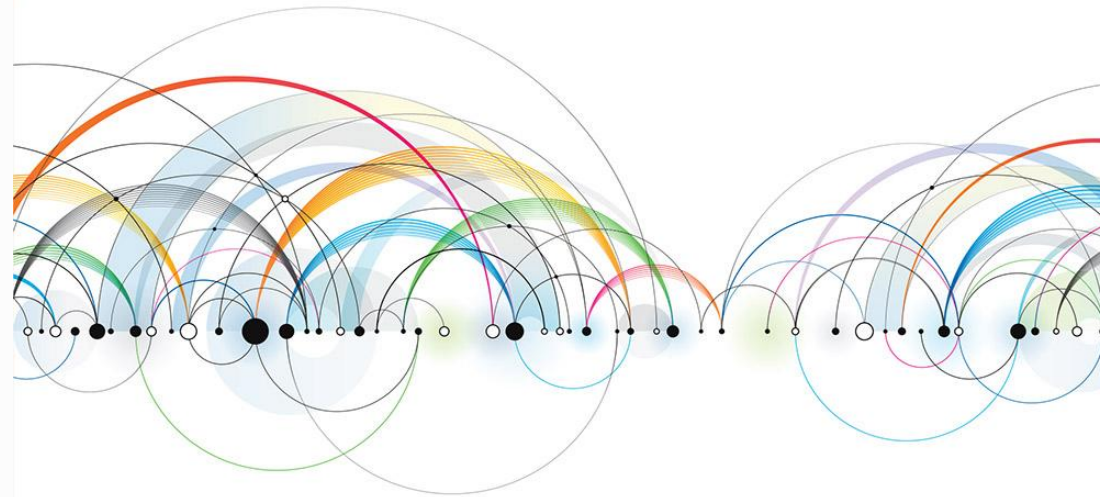
- **Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking**
- By Foster Provost and Tom Fawcett
- Amazon
 - https://www.amazon.com/Data-Science-Business-Data-Analytic-Thinking/dp/1449361323/ref=sr_1_1

"A must-read resource for anyone who is serious about embracing the opportunity of big data."

—Craig Vaughan, Global Vice President, SAP

Data Science *for* Business

What You Need to Know
About Data Mining and
Data-Analytic Thinking



Foster Provost & Tom Fawcett